

# Dynamics - Newton's Laws and Free-Body Diagrams

Course: Physics

## Core Ideas

Newton's First Law describes inertia. Newton's Second Law relates net force, mass, and acceleration:  $F_{\text{net}} = ma$ . Newton's Third Law states that interacting objects exert equal-magnitude, opposite-direction forces on one another.

## Free-Body Diagrams

Draw only forces acting ON the chosen object. Common forces include gravity, normal force, tension, friction, and applied force. Do not draw velocity or acceleration arrows as forces.

## Worked Example

A 10 kg box is pushed right with 50 N while friction acts left with 20 N. Net force = 30 N right. Using  $F_{\text{net}} = ma$ , acceleration =  $30/10 = 3.0 \text{ m/s}^2$  right.

## Inclines

For an object on an incline, gravity can be resolved into components parallel and perpendicular to the surface. The parallel component pulls the object down the slope.

## Common Error

The normal force is not always equal to  $mg$ . That shortcut only works in specific situations, such as a level surface with no other vertical forces or vertical acceleration.

## Practice

- A 4 kg object experiences a net force of 18 N east. Find its acceleration.
- Draw a free-body diagram for a book resting on a table.
- Explain the action-reaction pair when a swimmer pushes water backward.